



Unit IV: Intellectual Property

Intellectual Property Rights (IPR)

PPT PRESENTED BY

NAME : Srinivas Rao Tammireddy

ROLL NO : 257Y1D5805

YEAR : I Year I Semester

ACADEMIC YEAR : 2025 - 2026

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

What is Intellectual Property?

Intellectual Property (IP) refers to creations of the mind — such as inventions, literary and artistic works, designs, symbols, and brand names. IPR represents the legal rights that protect these creations.

The Purpose of IPR

- **Encourage Innovation:** Grants creators exclusive monopoly rights, allowing them to recover R&D costs and profit from their work.
- **Public Disclosure:** In exchange for legal protection, inventors must publish the details of their creation, expanding global scientific knowledge.
- **Economic Asset:** IP can be bought, sold, licensed, or used as collateral for corporate funding.
- **Consumer Protection:** Trademarks prevent customer confusion by verifying the origin of products.

Four Primary Types of IP

- **Patents:** Protects novel, useful inventions and processes (e.g. computer processors). Typically granted for 20 years.
- **Industrial Designs:** Protects the visual, aesthetic appearance of an object (e.g. smart phone shape).
- **Trademarks:** Protects brand names, logos, and slogans (e.g. 'Apple' leaf logo). Prevents brand copycats.
- **Copyrights:** Protects original artistic, musical, or literary works. E.g. books, music, and software code.

Inventions and Patentability Criteria

A patent is a government grant giving an inventor exclusive rights to exclude others from using the invention:

Three Criteria for Patentability

1. Novelty (Newness):

The invention must not be described in prior art anywhere in the world before the filing date.

2. Inventive Step (Non-Obviousness):

Must not be obvious to a person with ordinary skill in that technical field.

3. Industrial Applicability (Utility):

Must be useful and capable of being manufactured or used in some industry.

What CANNOT be Patented?

Most patent systems (including India's) exclude:

- **Mathematical Formulas:** Pure algorithms or mathematical models (e.g. Einstein's $E=mc^2$).
- **Laws of Nature:** Discovering a new plant or natural mineral in the wild.
- **Business Methods:** Abstract business concepts (e.g., standard auction steps).
- **Software as Code:** Raw code is protected by Copyrights, not patents. However, software that controls physical hardware can be patented in some regions.

Process of Patenting & PCT

How patents are filed locally and globally using international treaties:

Standard Patenting Timeline

- 1. Innovation & Prior Art Search:** Verify novelty using patent databases (Espacenet, USPTO).
- 2. Draft Specification:** Write the patent description and compile the precise 'claims' defining your invention's boundaries.
- 3. File Application:** File a provisional application to secure your 'priority date'.
- 4. Publication & Examination:** Patent office publishes the application and reviews claims against prior art.
- 5. Grant of Patent:** If approved, the patent is granted, requiring annual renewal fees.

Patent Cooperation Treaty (PCT)

- **Territorial Limitation:** Patents are national. A patent granted in India does not protect your invention in the USA.
- **The PCT Solution:** Administered by WIPO, the PCT allows inventors to file a SINGLE 'international' application to secure priority in 150+ member countries simultaneously.
- **Priority Window:** Secures a priority window of 30 months. This gives inventors time to search for foreign investors before paying expensive foreign filing fees.

Applications of IP Protection

Intellectual property rights are critical business assets across the tech sector:



Software Innovations Protection

Tech companies protect proprietary software engines (like search algorithms or video compression codecs) using patents, while protecting source code files using copyrights.



Industrial Design Rights

Automotive and consumer tech companies register visual designs (like smartphone cases, car body lines, or watch faces) to prevent competitors from mimicking their aesthetics.



Pharmaceutical R&D Patents

Drug companies patent new chemical compounds (typically costing billions to develop). Exclusive rights allow them to recover development costs prior to generic competition.



Standard Essential Patents (SEPs)

Telecommunications groups patent technologies essential to standards (like 5G). They license these to hardware makers on FRAND terms, earning royalties.



THANK YOU

Department of Computer Science and Engineering
Marri Laxman Reddy Institute of Technology and Management